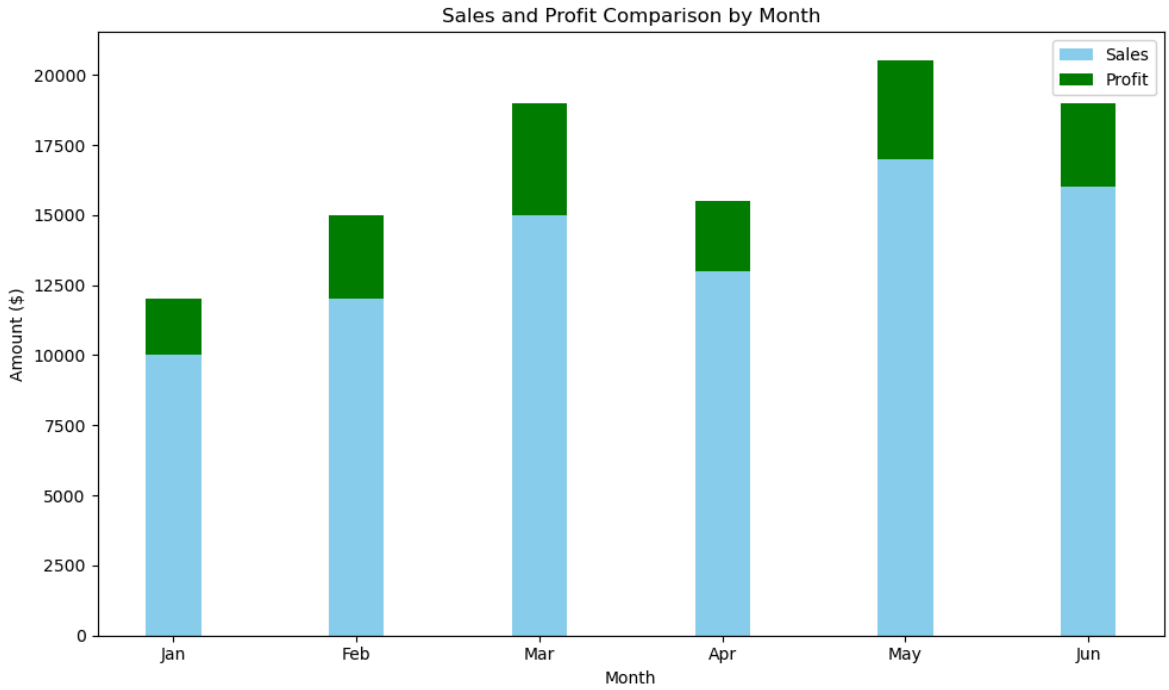
PANDAS

In [ ]:
```
# Raw data-the dataset which has missing value is called raw data.
# clean data- clean data or data cleansing
```

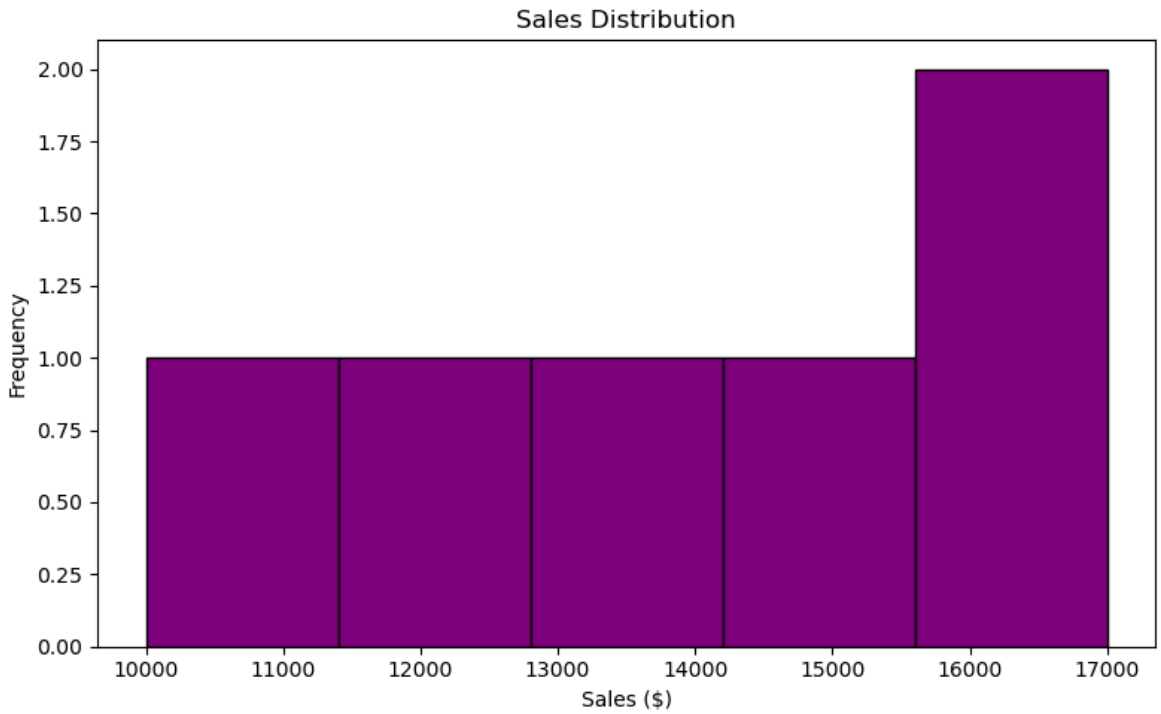
In [23]:
```
import pandas as pd
import matplotlib.pyplot as plt
# Sample data
data = {
    "Month": ["Jan", "Feb", "Mar", "Apr", "May", "Jun"],
    "Sales": [10000, 12000, 15000, 13000, 17000, 16000],
    "Profit": [2000, 3000, 4000, 2500, 3500, 3000]
}

df = pd.DataFrame(data)
```

In [3]:
```
# 1- line plot sales over time

import matplotlib.pyplot as plt

# Line Plot: Sales over Time
plt.figure(figsize=(8, 5))
plt.plot(df['Month'], df['Sales'], color='blue', marker='o', linestyle='-', labe
plt.title('Sales Trend Over Months')
plt.xlabel('Month')
plt.ylabel('Sales ($)')
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()

# Output: A line plot showing the sales trend over the months.
```

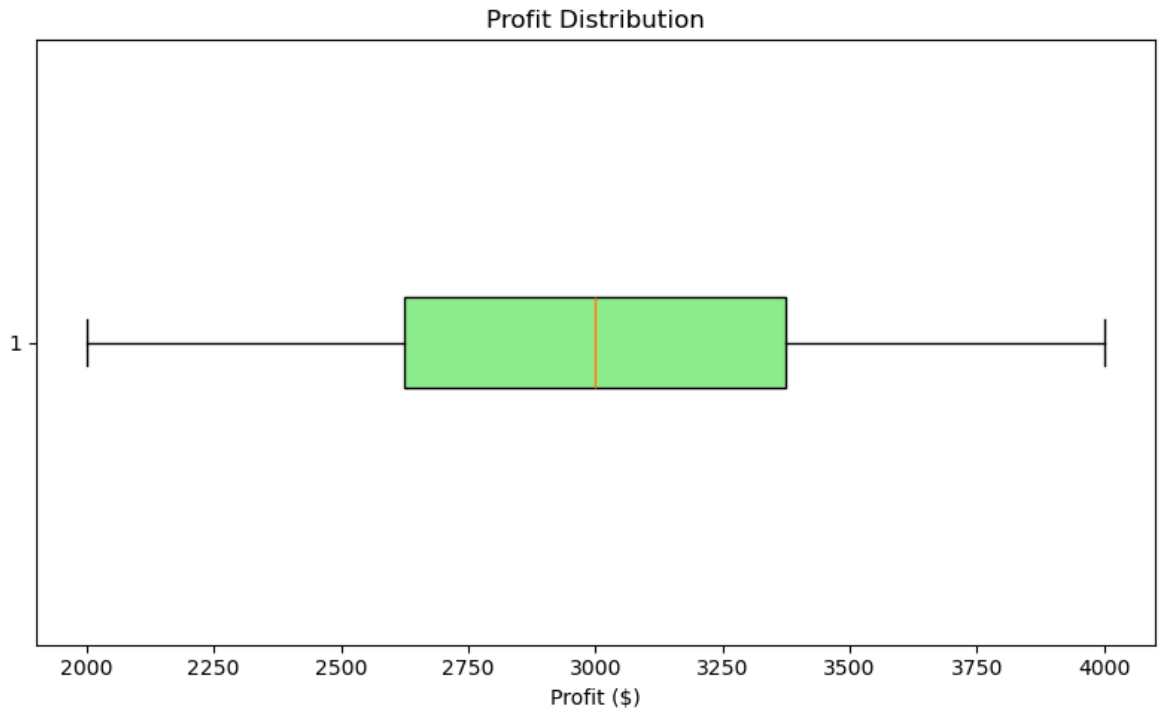
In [6]:
```
# Bar Plot: Sales vs Profit by Month
plt.figure(figsize=(10, 6))
width = 0.3
plt.bar(df['Month'], df['Sales'], width=width, label='Sales', color='skyblue')
plt.bar(df['Month'], df['Profit'], width=width, label='Profit', color='green', b
plt.title('Sales and Profit Comparison by Month')
plt.xlabel('Month')
plt.ylabel('Amount ($)')
plt.legend()
plt.tight_layout()
plt.show()

# A stacked bar plot where you can compare Sales and Profit for each month.
```



In [9]:
```
# Histogram: Distribution of Sales
plt.figure(figsize=(8, 5))
plt.hist(df['Sales'], bins=5, color='purple', edgecolor='black')
plt.title('Sales Distribution')
plt.xlabel('Sales ($)')
plt.ylabel('Frequency')
plt.tight_layout()
plt.show()

# A histogram showing the distribution of Sales values across months.
```

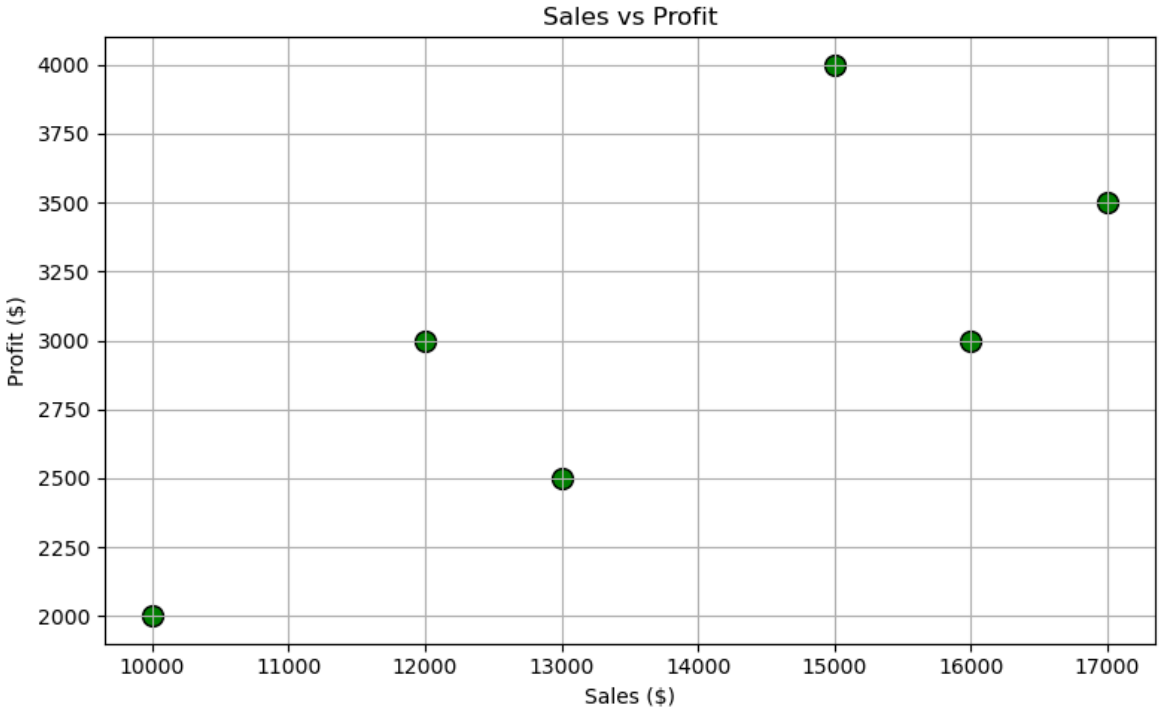


In [11]:
```
# Box Plot: Profit Distribution
plt.figure(figsize=(8, 5))
plt.boxplot(df['Profit'], vert=False, patch_artist=True, boxprops=dict(facecolor
plt.title('Profit Distribution')
plt.xlabel('Profit ($)')
plt.tight_layout()
plt.show()

# A box plot showing the spread and outliers of Profit.
```

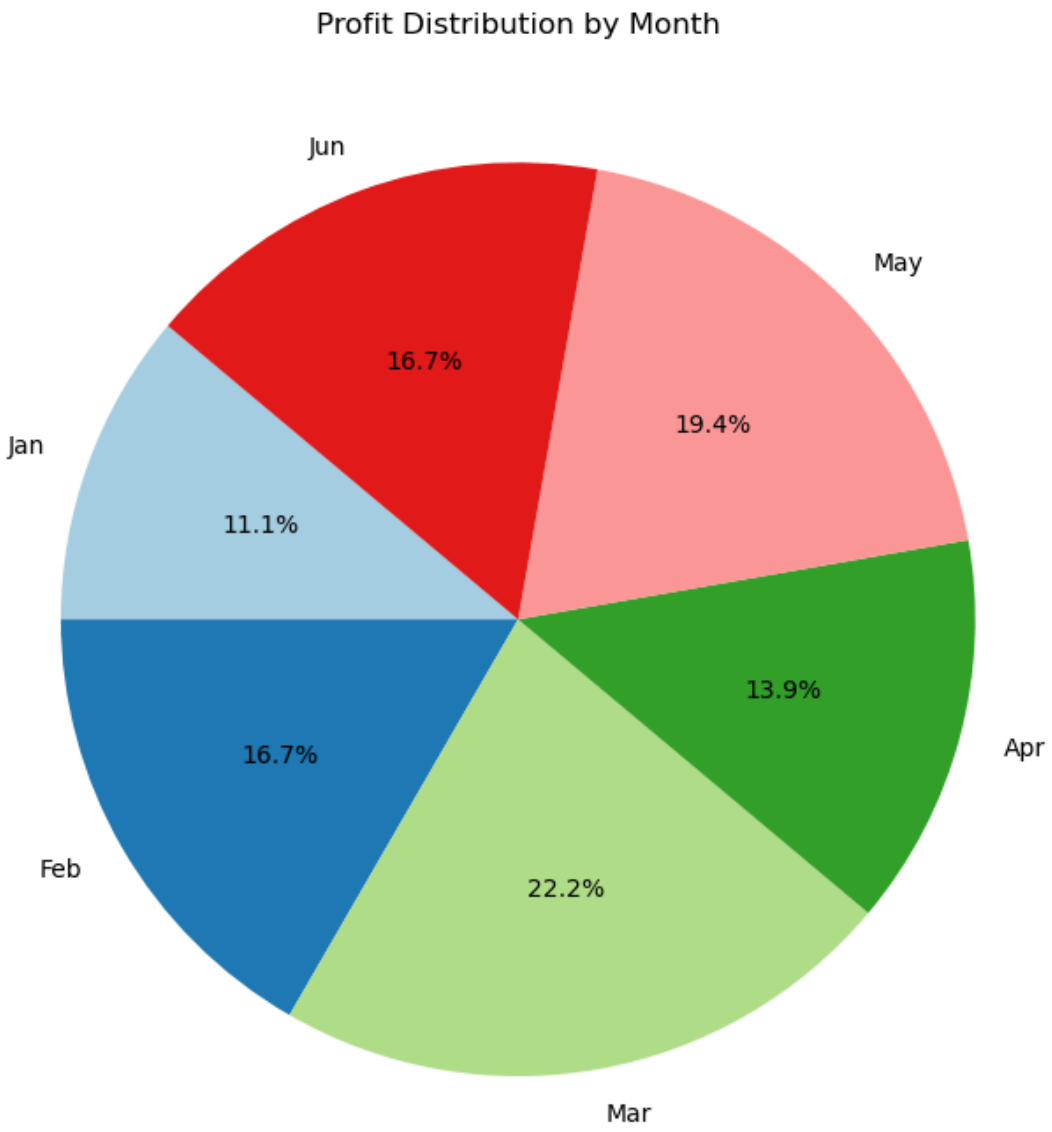


In [12]:
```
# Scatter Plot: Sales vs Profit (Correlation)
plt.figure(figsize=(8, 5))
plt.scatter(df['Sales'], df['Profit'], color='green', s=100, edgecolors='black')
plt.title('Sales vs Profit')
plt.xlabel('Sales ($)')
```

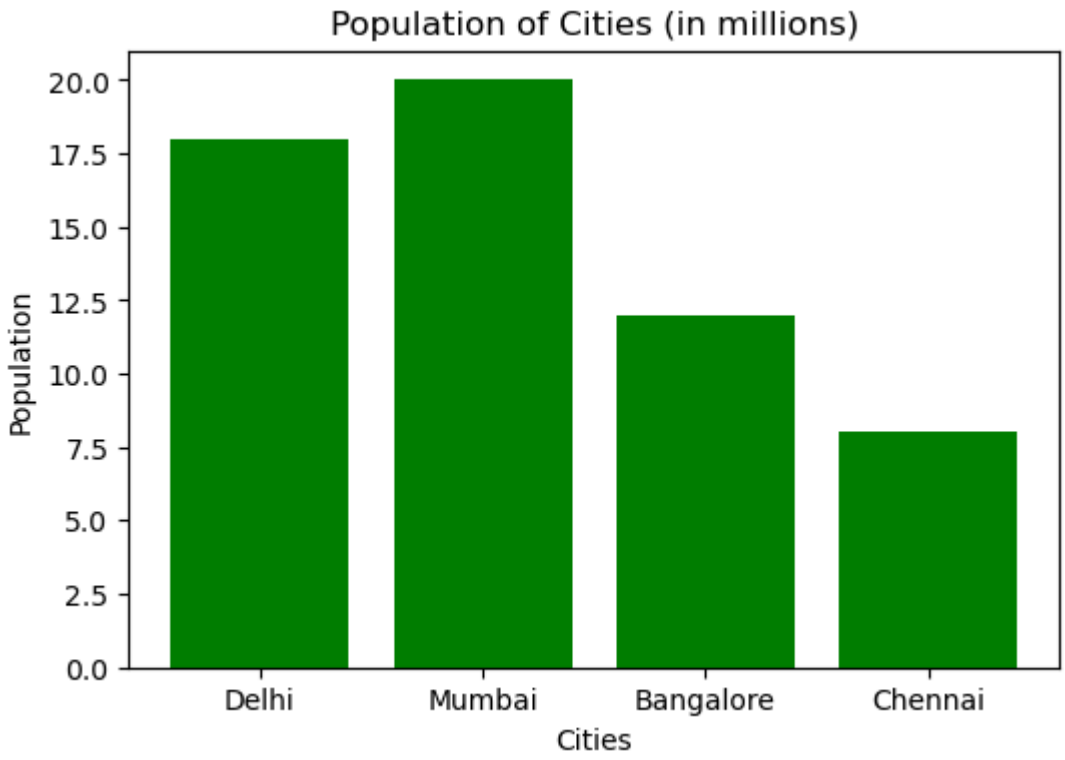
```
plt.ylabel('Profit ($)')
plt.grid(True)
plt.tight_layout()
plt.show()

# A scatter plot showing the correlation between Sales and Profit.
```



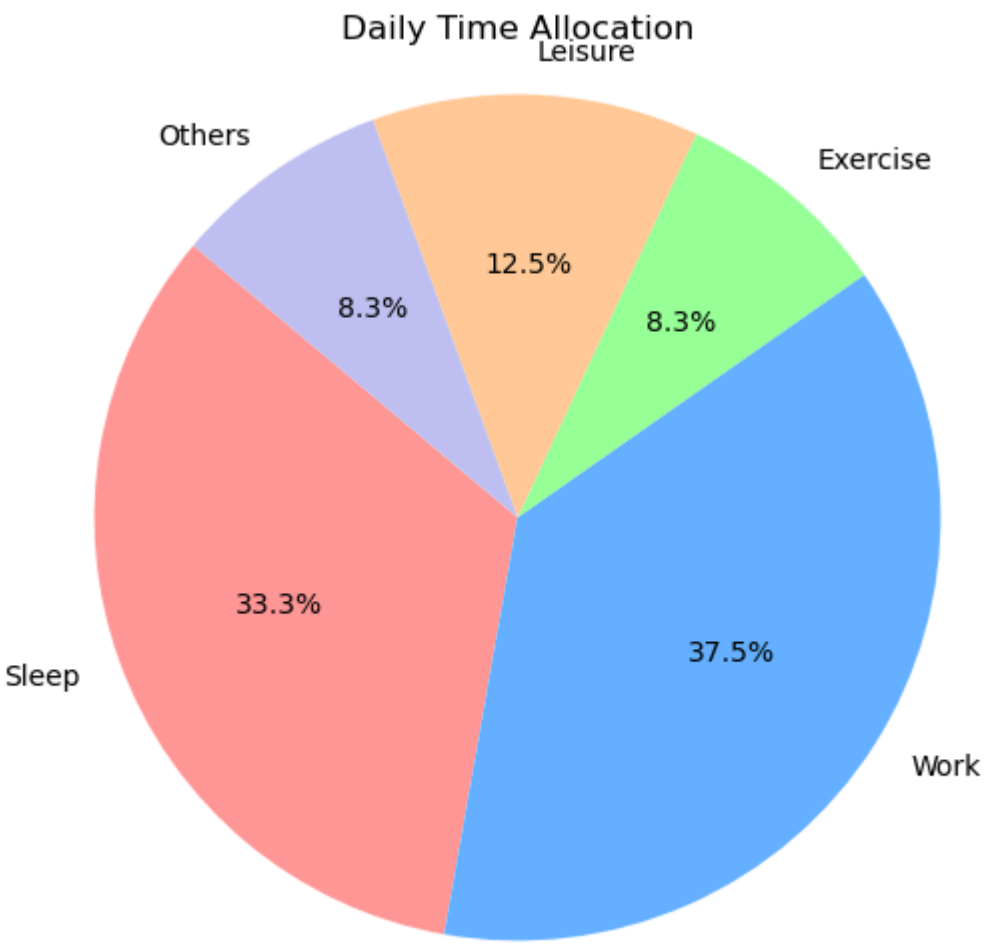
In [16]:
```
# Pie Chart: Profit Distribution by Month
plt.figure(figsize=(7, 7))
plt.pie(df['Profit'], labels=df['Month'], autopct='%1.1f%%', startangle=140, col
plt.title('Profit Distribution by Month')
plt.tight_layout()
plt.show()

# A pie chart displaying the proportion of profit distribution for each month.
```



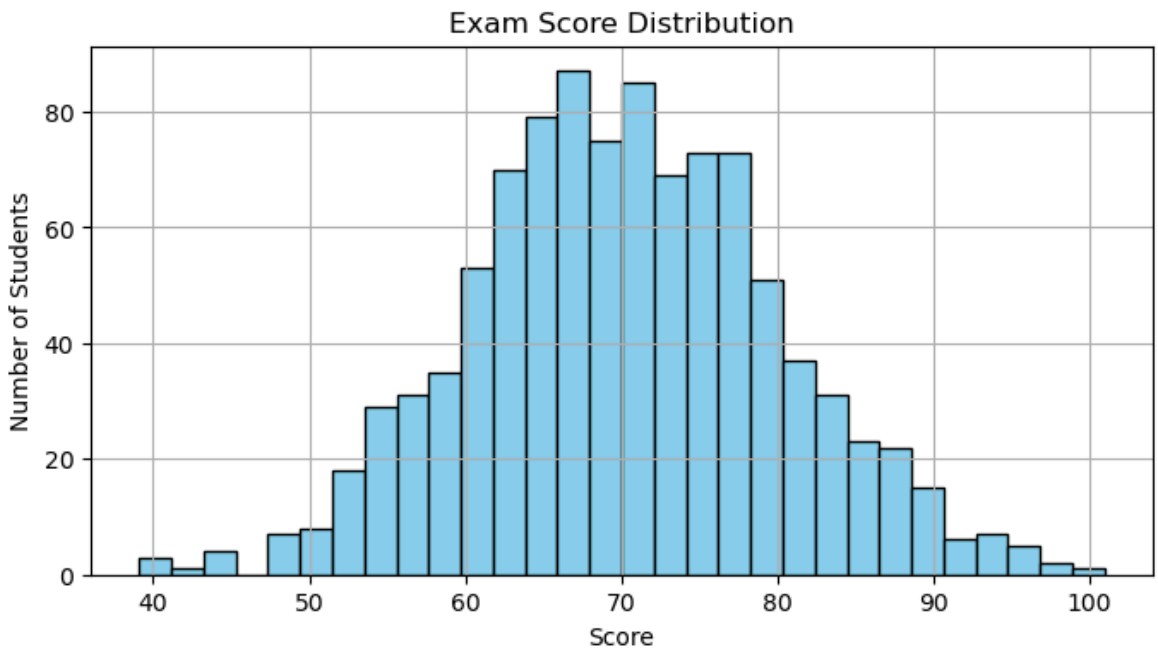
In [17]:
```
#bar chat

cities = ['Delhi', 'Mumbai', 'Bangalore', 'Chennai']
population = [18, 20, 12, 8]

plt.figure(figsize=(6,4))
plt.bar(cities, population, color='green')
plt.title("Population of Cities (in millions)")
plt.xlabel("Cities")
plt.ylabel("Population")
plt.show()
```



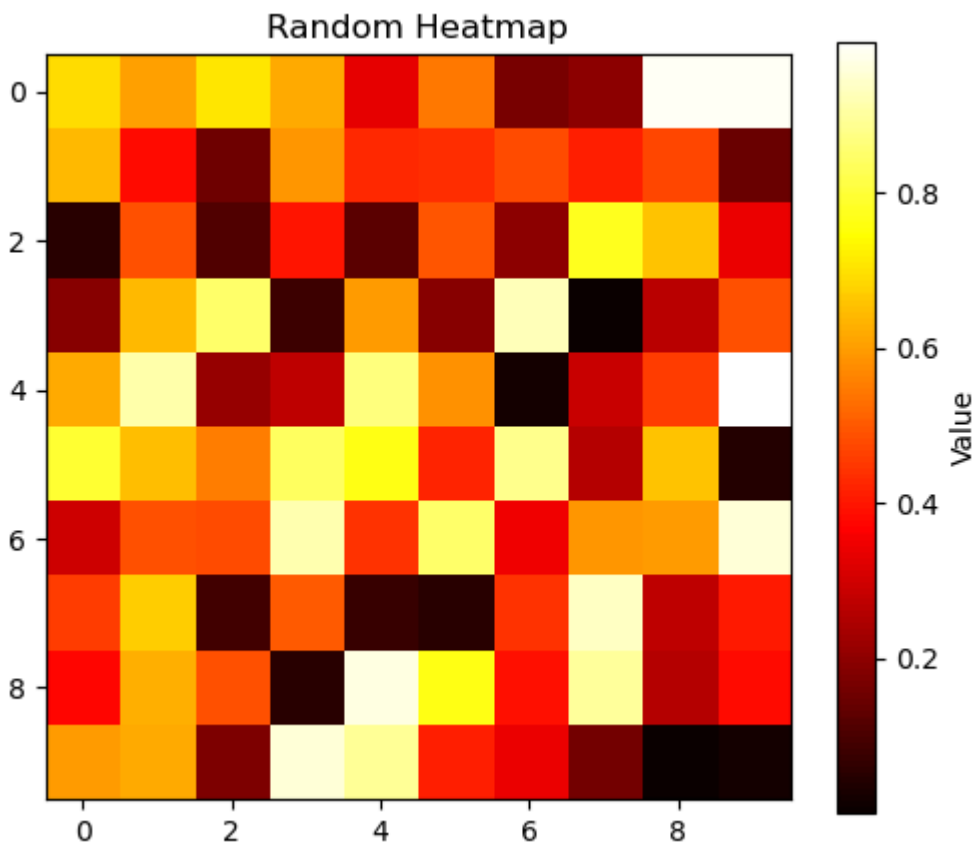
In [18]:
```
#pie chart
labels = ['Sleep', 'Work', 'Exercise', 'Leisure', 'Others']
sizes = [8, 9, 2, 3, 2]
colors = ['#ff9999','#66b3ff','#99ff99','#ffcc99','#c2c2f0']

plt.figure(figsize=(6,6))
plt.pie(sizes, labels=labels, colors=colors, autopct='%1.1f%%', startangle=140)
plt.title("Daily Time Allocation")
plt.axis('equal')  # Equal aspect ratio ensures that pie is drawn as a circle.
plt.show()
```



In [22]:
```
#histogarm
import numpy as np
scores = np.random.normal(loc=70, scale=10, size=1000)

plt.figure(figsize=(8,4))
plt.hist(scores, bins=30, color='skyblue', edgecolor='black')
plt.title("Exam Score Distribution")
plt.xlabel("Score")
plt.ylabel("Number of Students")
plt.grid(True)
plt.show()
```



In [21]:
```
#heatmap
import numpy as np
data = np.random.rand(10, 10)

plt.figure(figsize=(6, 5))
plt.imshow(data, cmap='hot', interpolation='nearest')
plt.colorbar(label='Value')
plt.title("Random Heatmap")
plt.show()
```



In [ ]: